

Unequal Job Opportunities and the Measurement of Welfare Inequality

Hisham Haydar

LISER and University of Luxembourg

France, EU-SILC priced through EUROMOD.

Seminar of 17 September. Welfare content is presented under ruling R6; DECOMP-2 is preliminary model-based accounting.

Thank you. The question is how much of the inequality in money-metric well-being is attributable to unequal job opportunities rather than to differences in taste, and what happens to that attribution when a model leaves opportunities out. I will show the model, the estimates, the fit, the welfare definition, a descriptive welfare baseline and the bounded DECOMP-2 accounting exercise. The final decomposition of total well-being inequality is not ready, and I will say exactly where that boundary lies.

Question and contribution

Two people with the same tastes and the same wage need not face the same set of jobs.

- How much of measured welfare inequality is attributable to unequal **job access** and **earning opportunities**, rather than to **preferences**?
- What does a model that assumes a common choice set do to that attribution?

Both questions are about a *measure*, not about a policy counterfactual.

The two questions are the same two questions as in every version of this paper; nothing in the recent rulings touched them. The first is an attribution question about a money-metric inequality index. The second is a methodological question: a discrete-choice labour supply model that gives every household the same alternatives has to explain non-employment with a taste parameter, and I want to know whether that relabelling survives into the welfare accounting.

The contribution is a structural opportunity set plus a welfare measure defined on it.

- A random-utility, random-opportunity (RURO) model in which *which jobs a household can reach* is estimated, not assumed.
- The Measure-1 money metric; in this empirical specification its direct reference collapses to universally available non-employment.
- A bounded three-factor exercise over systematic utility heterogeneity, coarse geographic/temporal access and earning opportunities.

The three-factor exercise is preliminary model-based accounting, not the final decomposition of total well-being inequality.

The contribution has three parts. Preferences and opportunity components are jointly estimated, with their separation relying on maintained functional-form and exclusion restrictions. The welfare measure is taken from the companion theory paper with Francois Maniquet. Within the bounded DECOMP-2 game, earning-opportunity heterogeneity has a larger Shapley contribution than the model's coarse geographic/temporal access channel. The final counterfactual-attainment estimand remains under design.

Data and the structural RURO model

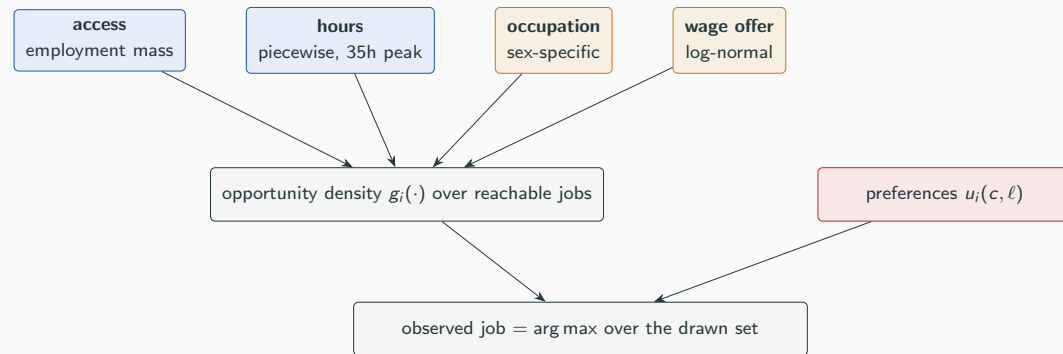
France, EU-SILC priced through EUROMOD: 1 540 single-adult and 2 223 couple households.

- Estimation frames: corrected criterion-A panels, S11 model of record.
- Estimation frame: each household's observed job plus 100 drawn alternatives, every one priced through the tax–benefit system.
- Fit diagnostics use a separate, larger panel: 2 048 common quadrature nodes per household.
- Weights are the survey weights dwt throughout.

S11 is the single model of record on these slides. Earlier S8 and R240 frames and their estimates are not shown and are not comparable.

Two estimation samples, singles and couples, on the corrected criterion-A frames. Everything on the slides after this point is S11. I am deliberately not showing the older S8 or R240 numbers alongside, because the frames differ and a side-by-side would invite a comparison the samples do not support. Each household contributes its observed job and a hundred drawn alternatives, each of which is run through EUROMOD so the consumption argument is a real disposable income, not an approximation. The second number on the slide is a different object: the positive-fit diagnostics (POSFIT v2b) integrate over a common panel of two thousand and forty-eight priced quadrature nodes per household, not over the estimation draws.

A job is a package; a household draws from its own distribution over packages.



Access and capability are not separately identified; the access factor is their product.

This is the architecture. A job is not an hours cell: it is hours, occupation and a wage, and whether the household can reach it at all. The opportunity density factorises into an access margin, a piecewise-constant hours factor with a peak at the statutory thirty-five hour week, sex-specific occupation availability, and an occupation-conditioned log-normal wage offer. Preferences enter separately. The observed job is the argmax over what was drawn. The identification caveat is on the slide and I keep it there all talk: the access factor mixes personal capability and market availability and the cross-section does not separate them.

Preferences and the opportunity density are estimated in one likelihood.

$$\Pr(j_i | x_i) \propto \exp\{ u_i(c_{ij}, \ell_{ij}) + \log g_i(j) - \log q_i(j) \}$$

u preferences g estimated opportunity density q numerical proposal density, divided out

The proposal correction is what makes the drawn alternatives a simulator for the estimated set rather than the choice set itself.

One likelihood, two structural objects. The proposal density is a numerical device: the alternatives are drawn from it and it is divided back out, so it is not part of the economics. That separation matters later, because the welfare measure I am going to define must not depend on the proposal density, and I will show that it does not. Preferences and opportunity components are jointly estimated, with their separation relying on maintained functional-form and exclusion restrictions. No causal interpretation follows from that separation. The baseline deliberately keeps systematic preference heterogeneity parsimonious: age profiles for all four adult groups and a child-related shifter for women. The child term is interpreted as a reduced-form behavioural/time-constraint shifter, not as pure taste. More explicitly, systematic leisure heterogeneity contains an intercept, age, age squared and group-specific leisure curvature for all four groups, plus a child shifter for women; sex and household type are carried by separate parameter blocks.

Positive estimates and fit

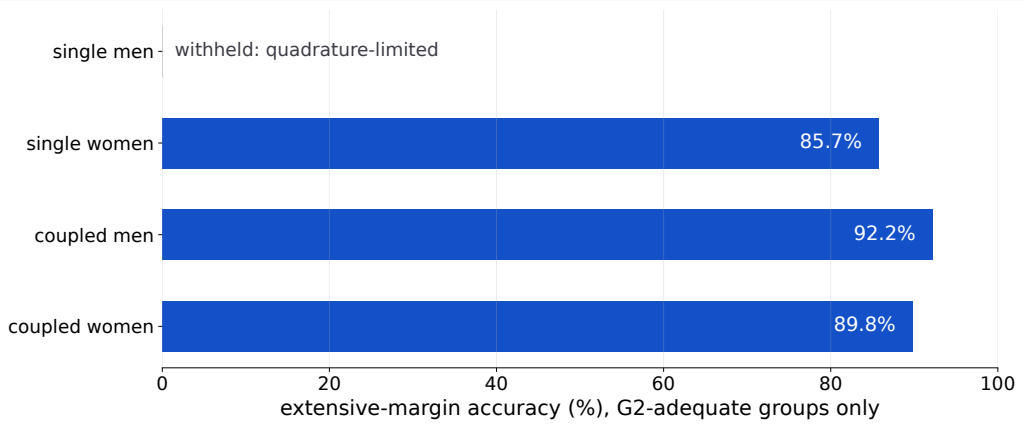
S11 model of record: 41 free parameters for singles, 47 for couples.

	singles	couples
free parameters	41	47
estimation households	1 540	2 223
objective at $\hat{\theta}$	6253.463	10283.034

Objectives reproduce exactly under the independent diagnostics evaluator, package commit 55bb0d0.

The headline fact about the estimates is that they are reproducible: an evaluator built separately for the diagnostics work recovers both stored objective values to the last digit shown. That is the precondition for everything on the next two slides. The parameter counts differ because the couples specification pins nothing at the bound while the singles specification pins eleven coefficients by construction.

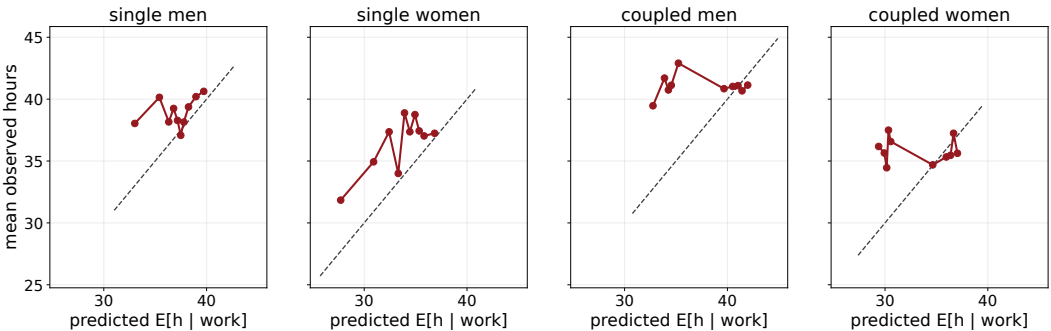
Extensive-margin accuracy, shown only for the groups where it passes the quadrature-adequacy gate.



Values are shown only where the G2 weighted quadrature gate clears for this statistic; single men are quadrature-limited and withheld. This is not a group-count result.

The pre-registered adequacy gate asks whether numerical integration error is small relative to sampling variation. I show values only where that specific extensive-accuracy statistic clears the gate and withhold the single-men value because it is quadrature-limited. This is not a group-count result. Aggregate participation and occupation margins are reproduced reasonably closely, while hours fit is uneven. Individual-level predictive diagnostics raise possible excess stochastic dispersion in some groups; the group-level classification is not used as a headline result. If asked: coupled men pass the weighted gate narrowly (ratio 0.240 against 0.25) and miss the unweighted one (0.250); the deck applies the weighted rule throughout. Source: POSFIT v3 (MNL_posfit, branch diagnostics/posfit-v3, commit 96693269); the extensive-margin numbers are unchanged from v2b (commit a2e80a8).

Calibration conditioned on prediction: observed hours against predicted hours, by predicted decile.

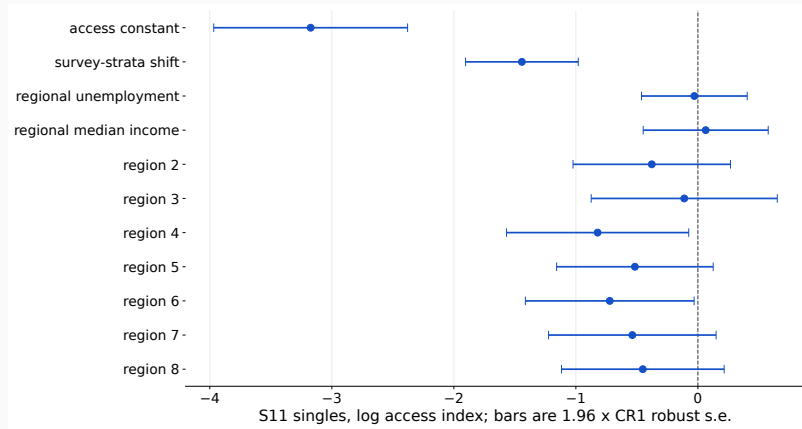


support coverage audited (POSFIT v2b); calibration statistics partly quadrature-limited; fit verdicts open pending Deputy review

Households are sorted into deciles of their own predicted expected hours conditional on working, and each decile’s mean observed hours is plotted against its mean prediction, on the v2b common-node support. The dashed line is the forty-five degree line. Conditioning on the prediction rather than on the outcome is the direction that does not mechanically manufacture regression to the mean. The caption is the honest status. The support audit has now been done: v2b re-evaluates the fit on a common quadrature panel far larger than the estimation draws, no household has an effective sample size below thirty, singles have no unsupported observed bin, and only a handful of couples fall in a region that finite panel does not represent. The calibration statistics behind this picture remain partly quadrature-limited, and the group-level classification is not used as a headline result. Aggregate participation and occupation margins are reproduced reasonably closely, while hours fit is uneven. So read the shape, not the gaps.

Evidence for heterogeneous opportunities

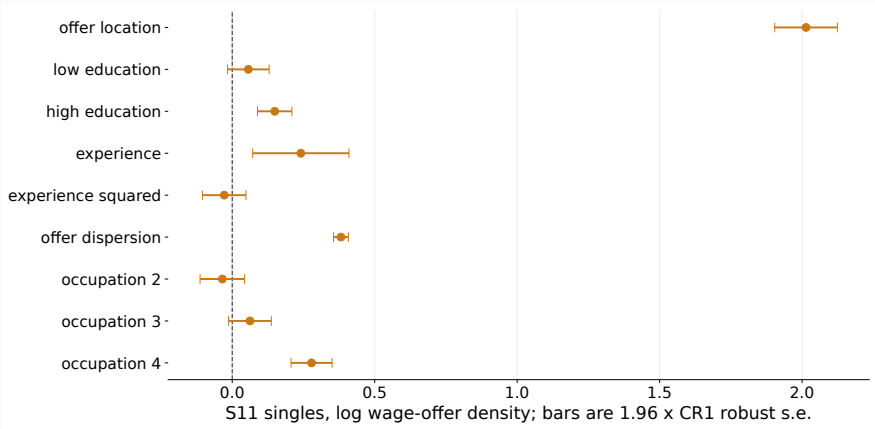
The access kernel is estimated, and it varies across households.



S11 singles; horizontal bars are $1.96 \times \text{CR1}$ cluster-robust standard errors. Structural and model-conditional; not causal.

This is the access block of the estimated opportunity density. The point is not any individual coefficient, it is that the block is estimated rather than imposed and that it moves: households in different regional access environments, and in different survey strata, face materially different employment masses at the same preferences and the same wage. This is the evidence that the opportunity side of the model is doing work. It is also the evidence the welfare measure later rests on, which is why I show it before the measure and not after.

So is the earning-opportunity kernel: an offer distribution, not a single wage.



S11 singles; occupation terms shift the offer location. Dispersion $\hat{\sigma} = 0.382$ ($z = 28.7$).

The wage side is an offer density with an estimated dispersion, not a point wage attached to each person. Education and experience shift the location, occupation shifts it again, and the residual dispersion is precisely estimated. Two households with identical observed wages can have different earning opportunities in this model, because what they face is a distribution. That is the second half of what I mean by unequal job opportunities, and together with the access block it is what the welfare measure has to be defined on.

The welfare measure

Haydar–Maniquet W^1 : the consumption at home that would leave the household exactly as well off.

$$W_i^1 = m_i(o; z_i) \quad \text{where} \quad u_i(m_i(o), o) = u_i(z_i)$$

z_i attained bundle o the non-employment state m_i the money metric

Mapping F is the primary empirical implementation (ruling R1). Mapping M is sensitivity only and requires an unidentified intensity parameter; it is not in the baseline.

Q&A note — ruling R1, JMP_W1_fork_ruling_v1.md Appendix A. If asked why this reference and not the market-job reference: the fork analysis compared Mapping F, the full feasible set including non-employment, against Mapping M, the market-job universe. R1 rules F primary for the current paper. M at its dense-law endpoint stops being a household-specific object and becomes a whole-market reference, and at finite intensity it needs a set-size parameter the data do not identify. That is recorded as a limitation and as future work, not as a competing headline. I am not claiming F is the right reference in the theory; I am claiming it is the one this empirical specification identifies.

Under the current specification W^1 coincides with the staying-home equivalent — as a fact about this domain.

$$W_{i,F}^1 = C_i^{\text{obs}} \exp \left[\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right]$$

- Inputs: C_i^{obs} , $L_i(j_i^{\text{obs}})$, $L_i(o)$, β_c — nothing else.
- No opportunity density g , no proposal q , no shock ε , no intensity κ .
- Current nonworkers: $W = C$; current workers: $W < C$ under the maintained empirical domain.

This is an **empirical-domain coincidence** under log consumption and universally available non-employment. It is **not** the theorem $W^1 = W^4$ in the Haydar–Maniquet theory.

Q&A note — ruling R1 and BASELINE-F-1 ruling section 2. The most likely question here is whether I have just proved that measure one and measure four are the same thing. I have not, and the ruling is explicit about it. The coincidence has two ingredients, both specific to what is estimated: consumption enters utility in logs, and non-employment is behaviourally available to every household. Change either and the two measures separate. If asked about the consumption argument: ruling E2 confirms that C^{obs} must be the exact object entering the estimated utility at the observed state, not a re-chosen normative income concept, and the measure-correspondence audit closed that provenance before the baseline was allowed to run. The money metric is derived from the Measure-1 reference-set principle. Under the current empirical specification its direct reference collapses to the universally available non-employment state; opportunity heterogeneity therefore affects the current welfare measure through attained bundles. For a one-nat shortfall, $L_i(j_i^{\text{obs}}) - L_i(o) = -1$, the illustration is $W = C_i^{\text{obs}} \exp(-1/\beta_c) < C_i^{\text{obs}}$.

**Equivalised W^1 -F distribution
(primary)**

Baseline W^1 -F, equivalised, single adults.

	C^{eq}	$W_F^{1\text{eq}}$
households	1 540	1 540
weighted mean	1 766	1 302
weighted median	1 588	1 165
weighted Gini	0.263292	0.249807

Units: EUR/month, household-equivalised (modified-OECD scale); survey weight dwt. A descriptive dispersion comparison, not a welfare-loss index. Equivalence scale: the ratified modified-OECD scale (Deputy ruling “SCALE CLOSED; CHILD-SHIFTER FRAMING”, s1; JMP_SCALE_REVIEW_1_equivalence_scale_economics_v1.md). Singles and couples equivalised levels are never compared.

Source: docs/results/ JMP_BASELINE_F1_equivalised_reporting_v1.md, commit 4c4e07e, over the verified construction (MNL commit 6048c9f, independently verified b5550af). This is a descriptive dispersion comparison (C^{eq} against $W_F^{1\text{eq}}$ within the singles sample), not a welfare-loss index.

Q&A note — BASELINE-F-1 ruling, E3-EQ, and ruling R6. Equivalised results are primary on this slide: the household-equivalised Gini of W_F^1 is below the household-equivalised Gini of observed consumption, for singles. I show that as a descriptive dispersion comparison only, not as a welfare-loss statement — the ratio W_F^1/C^{obs} reflects the systematic leisure term at the observed job relative to home leisure, not a monetary loss, and I am not constructing a loss index on this slide. The equivalence scale is the modified-OECD scale, ratified as the primary scale for current JMP distributional reporting by the Deputy ruling “SCALE CLOSED; CHILD-SHIFTER FRAMING”, section 1, on the economics review in JMP_SCALE_REVIEW_1_equivalence_scale_economics_v1.md; the scale question is closed. The unequivalised construction this equivalises is unchanged and is in the backup appendix.

Baseline W^1 -F, equivalised, couples.

	C^{eq}	W_F^{1eq}
households	2 223	2 223
weighted mean	2 245	1 311
weighted median	2 063	1 237
weighted Gini	0.226805	0.197403

Units: EUR/month, household-equivalised (modified-OECD scale); survey weight dwt. A descriptive dispersion comparison, not a welfare-loss index. Equivalence scale: the ratified modified-OECD scale (Deputy ruling “SCALE CLOSED; CHILD-SHIFTER FRAMING”, s1; JMP_SCALE_REVIEW_1_equivalence_scale_economics_v1.md). Singles and couples equivalised levels are never compared.

Reported separately from singles: no pooled figure, and no cross-sample level comparison is drawn on this slide or elsewhere in this deck. Source: same memo and commits as the singles equivalised slide.

Q&A note — BASELINE-F-1 ruling, E3-EQ, and ruling R6. Same descriptive comparison, on the couples sample. I am deliberately not placing this table next to the singles one: the two are separate estimation frames, the welfare unit differs, and R6 forbids presenting a singles/couples equivalised-level comparison as a finding. If asked how the couples numbers compare to singles, the honest answer is that this deck does not draw that comparison, and I would want to see it examined outside a seminar slide before treating it as informative. The scale is the same ratified modified-OECD scale as on the singles slide (Deputy ruling “SCALE CLOSED; CHILD-SHIFTER FRAMING”, section 1); ratifying the scale does not license a cross-sample level comparison.

Decomposition architecture

DECOMP-2 equalises three bounded channels, not whole job access.

P systematic utility heterogeneity (tastes + reduced-form time constraints) → common reference

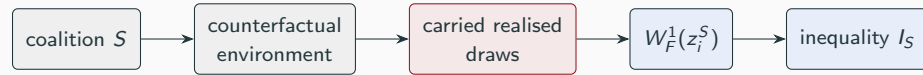
A local geographic/temporal access shifters (region, urban/rural, year) → common reference

B earning opportunities → common reference offer distribution

Household resources, needs and composition are held fixed. Personal occupation access, hours-band access and node-level alternative characteristics are not equalised by *A*.

Q&A note — ruling R6 and final economics review M5. For DECOMP-2, *A* is local geographic/temporal access shifters (region, urban/rural, year). It is not personal occupation access, hours-band access or full job access. Children enter the current structural specification through a reduced-form female time-constraint shifter. In a unitary labour-supply model this term may capture both tastes and childcare/home-production constraints; we therefore do not interpret it as pure preference heterogeneity. The baseline deliberately keeps systematic preference heterogeneity parsimonious: age profiles for all four adult groups and a child-related shifter for women. The child term is interpreted as a reduced-form behavioural/time-constraint shifter, not as pure taste.

DECOMP-2 is executed; the final attainment estimand remains under design.



bounded DECOMP-2 route

DECOMP-2 reuses the already-priced estimation panel with no re-estimation and no new pricing. The final decomposition's counterfactual-attainment estimand remains under design (R5).

Q&A note — ruling R5 and final economics review M10. The preliminary exercise carries the household's realised draws through each counterfactual environment. That is an executed, bounded accounting route, not the final counterfactual-attainment estimand. The final design still has to choose between a realised-bundle route with conditioned latent heterogeneity and an ex-ante g -computation route; the inequality of expected welfare is not the expected inequality of welfare.

Preliminary decomposition and boundary

Earning opportunities exceed coarse access within the bounded game.

In a preliminary three-factor structural exercise that holds household resources, needs and composition fixed, equalising systematic utility heterogeneity, coarse geographic/temporal access heterogeneity and earning opportunities changes money-metric well-being inequality by 1.8–9.9% of the baseline Gini, depending on household type and reporting scale. Within the Shapley allocation of that movable component, earning-opportunity heterogeneity has a larger contribution than the coarse geographic/temporal access channel in both samples and both reporting conventions. The preference contribution is not sign-robust to equalisation.

These are preliminary model-based accounting results, not causal estimates and not the final decomposition of total well-being inequality.

Q&A note — ruling R6 and final economics review M3–M11. The decomposition is bounded by design because household resources, needs and composition are held fixed. Within that bounded game, equalising P/A/B changes the Gini by 1.8–9.9% of its baseline level, depending on sample and reporting scale. The restriction is by construction; the realised numerical magnitude is a model result under that restriction. Dispersion in consumption is quantitatively large relative to dispersion in log well-being: $\text{Var}(\log C)$ is roughly 100–127% of $\text{Var}(\log W)$, with the excess offset by a large negative covariance between consumption and the leisure valuation. DECOMP-2 holds household resources, needs and composition fixed, leaving important sources of dispersion outside the P/A/B allocation. If asked about wage elasticities: a valid gross-wage perturbation requires new tax-benefit repricing over the affected job alternatives, and the current priced support does not contain that counterfactual.

What I would like from you today.

- Is Mapping F the reference you would want for this question?
- Which counterfactual attainment estimand would you defend?
- What would convince you the access kernel is identified, given that access and capability are not separated?

Thank you.

Q&A note — rulings R1, R5, and the identification caveat carried from the model slide. Three questions, in the order I most need answers. The reference domain choice is normative and I would like it challenged. The attainment estimand is the blocking design decision. And the access identification caveat is the one every discussant raises; I would rather hear what evidence would settle it than defend the current position.

households	1 540
weighted mean	1 435
weighted median	1 320
weighted Gini	0.238
observed workers	1 336
observed non-workers	204

Units: household EUR/month, unequivalised; survey weight dwt. Descriptive baseline; no counterfactual, no decomposition, no causal reading.

Secondary/verification slide: the equivalised singles slide (§6) is primary. Source: MNL/outputs/welfare/baseline_f1_v1/ baseline_f1_full_sample_report_v1.md, commit 6048c9f, independently verified b5550af.

This is the literal observed-bundle Mapping-F distribution before equivalisation, kept here so any question about the un-scaled level or about worker/non-worker counts has a one-slide answer. It is not the headline: the equivalised slide in §6 is. For non-workers the measure equals observed consumption exactly by construction; for workers it is strictly below it.

households	2 223
weighted mean	2 496
weighted median	2 356
weighted Gini	0.210
observed workers	2 173
observed non-workers	50

Units: household EUR/month, unequivalised; survey weight dwt. Descriptive baseline; no counterfactual, no decomposition, no causal reading.

Secondary/verification slide: the equivalised couples slide (§6) is primary. Reported separately from singles. No pooled figure: the two samples are separate estimation frames and the welfare unit differs.

The couples backup is on its own slide and there is no pooled row: the two samples are separate estimation frames with different welfare units, and even on the ratified modified-OECD scale this deck never compares singles and couples levels. These are unequivalised household totals: couples pool two earnings streams, and the non-employment count is small. The pricing-floor issue that affects the neither-work bundle for a subset of couples does not enter here, because this baseline is evaluated at the observed bundle and never prices the home state.

check	singles	couples
C1 non-workers, $\max W^1 - C^{\text{obs}} $	0	0
C2 workers, ratio outside $(0, 1)$	0	0
C3 dependency test $(g, q, \varepsilon, \kappa)$	PASS	PASS
C4 MEASURE-MAP-1R worked households	PASS	PASS
C5 indifference identity, $\max \Delta u $	1.8e-15	1.8e-15
C6 guards triggered	0	0

All six checks were pre-registered in the BASELINE-F-1 ruling before execution. C2 violations were to be explained, never clipped; none occurred.

The checks are the reason the baseline is presentable. C1 and C2 are the model-implied domain conditions. C3 is the dependency test: the baseline API accepts no opportunity density, no proposal, no shock and no intensity, and does not read them indirectly. C5 inverts the utility numerically and confirms the indifference identity to machine precision.

group	extensive accuracy	G2 label
coupled women	89.8%	ADEQUATE (ratio 0.02)
single women	85.7%	ADEQUATE (ratio 0.15)
coupled men	92.2%	ADEQUATE (ratio 0.24)
single men	—	QUADRATURE-LIMITED, withheld

Weighted, all households. Ratio is Monte Carlo s.e. over sampling s.d.; the gate is ≤ 0.25 . Node bootstrap: 200 replicates over 2 048 common quadrature nodes per household (diagnostic panel, not the estimation draws). Simulated bands: 500 outcome vectors at fixed $\hat{\theta}$. POSFIT v2b, commit a2e80a8.

The gate was fixed before the statistics were computed. It asks whether numerical integration error is small enough relative to sampling variation for the statistic to be a statement about the model rather than about the quadrature. Withholding rather than downgrading is deliberate: a quadrature-limited number carries no information about fit.

ruling	what it settles here
R1	Mapping F primary; $W^1 = W^4$ is a domain coincidence
R2	W1-EA retired from the baseline; preserved, not repaired
R4	MEASURE-MAP-1R: the measure-correspondence gate
R5	counterfactual attainment: in design, not executed
R6	what this seminar may present
E1	BASELINE-F-1 authorised, executed, independently verified
E2	C^{obs} is the estimated utility's own argument
E3	C^{obs} benchmark: not authorised, not produced
E3-EQ	equivalised reporting (§6) primary; modified-OECD ratified (SCALE CLOSED, s1)

R1–R6: docs/normative/JMP_W1_fork_ruling_v1.md, Appendix A. E1–E3: docs/normative/JMP_W1_BASELINE_F1_authorization_and_Cobs_ruling_v1.md. E3-EQ: docs/results/JMP_BASELINE_F1_equivalised_reporting_v1.md, commit 4c4e07e.

This backup slide exists so that any question of the form “why is that not here” has a one-line answer with a document behind it.